

BONAM MINGOLE

✉ bjm6940@psu.edu ♦  [bonam-m.github.io](https://github.com/bonam-m) ♦  github.com/Bonam-M ♦  linkedin.com/in/bonammingole

Ph.D. Student in Informatics ♦ Penn State University ♦ State College, PA, USA

EDUCATION

Pennsylvania State University (GPA: 3.90/4.0)

Ph.D. in Informatics

Advisor: Prof. Amulya Yadav | *Research focus*: Socially Responsible AI

State College, USA

Aug. 2024 - Present

University of Quebec in Montreal (GPA: 3.67/4.0)

M.S. in Software Engineering

Highest Honors Thesis: Analysis and co-creation of an AI assistant in the Financial Industry.

Montreal, Canada

Sept. 2017 - Jan. 2020

TECHNICAL SKILLS

Programming Languages

Python, SQL, Java, CSharp, PHP, HTML, Bash, LaTeX

Framework & Tools

Git, Pytorch, Sci-Kit Learn, LangChain, Huggingface Transformers

PUBLICATIONS

6. **Mingole, B.**, Yadav, A. (2025). *Evaluating Generative AI diagnostic ability and potential harms to the average user's health*. In progress.  *Media Coverage*: [Penn State News](#)
5. Rismani, S., Davis, L., **Mingole, B.**, Rostamzadeh, N., Shelby, R., Moon, A. (2025) *Responsible AI Measures Dataset: A Dataset of Ethics Evaluation of AI Systems*. Under review - Nature Scientific Data 2025.
4. Guo, H., Venkit, P. N., Jang, E., Srinath, M., Zhang, W., **Mingole, B.**, Gupta, V., Varshney, K.R., Sundar, S.S. and Yadav, A. (2024). *Hey GPT, Can You be More Racist? Analysis from Crowdsourced Attempts to Elicit Biased Content from Generative AI*. arXiv preprint. arXiv:2410.15467. Under review - ACL 2025.
3. Wang, X., Zhang, W., Koneru, S., Guo, H., **Mingole, B.**, Sundar, S.S., Rajtmajer, S. and Yadav, A. (2024). *The Reopening of Pandora's Box: Analyzing the Role of LLMs in the Evolving Battle Against AI-Generated Fake News*. arXiv preprint arXiv:2410.19250. Accepted - NAACL 2025.
2. **Mingole, B.**, Gilmour, A., Diallo, D., Patel, J., Mussio, M. (2023). *AITHICAL: an AI Ethics guidelines aggregator Prototype*. AI4Good Lab 2023 Conference.
 *Media Coverage*: [AI4Good Lab Projects](#)
1. **Mingole, B.** (2020). Analysis and Co-creation of an AI assistant in the financial industry [Master's thesis, University of Quebec in Montreal]. Archipel UQAM. <https://archipel.uqam.ca/15103/1/R1002.pdf>

RESEARCH & INDUSTRY EXPERIENCE

Graduate Research Assistant | Penn State University (State College, USA)

Aug. 2024 - Now

Research Topic: Responsible AI — *Advisor*: Prof. Amulya Yadav

- Conducted studies to assess Generative AI pitfalls, including biases, misinformation, and [health misdiagnoses](#).
- Built a Retrieval Augmented Generation (RAG) pipeline using medical textbooks to enhance Large Language Models, diagnostic ability, achieving ~80% accuracy.
- Evaluated a real-world NLP model fairness by comparing the F1 score between geographical regions.
- Leveraged *Latent Dirichlet Allocation* for topics extraction from *code-mix* SMS messages (English and Swahili).

Responsible AI Research Assistant | Mila Quebec AI Institute (Montreal, Canada)

Aug. 2023 - Aug. 2024

- Conducted a *Scoping Review* on how AI researchers operationalize ethical principles through measurement.
- Extracted data from 257 papers (from 2011 - 2023) measuring AI systems adherence to Responsible AI principles.
- Created the [Responsible AI Measures Dataset](#), consolidating 11,865 data points across 791 evaluation measures, covering 11 ethical principles including Fairness, Transparency, Privacy, Trust, Sustainability, and more.
- Developed a taxonomy of metrics and evaluation methods for Responsible AI principles.

Research Review Coordinator [Part-time position] | Mitacs (Montreal, Canada) *Jan. 2022 - Aug. 2024*

- Evaluated 10,000+ AI research grant proposals, contributing to the allocation of \$50M+ in research funding.
- Implemented a structured evaluation framework, cutting review time by 20% while maintaining quality.
- Bridged collaboration among academia, industry, and nonprofits, enabling 130+ joint initiatives.
- Analyzed proposals trends to identify emerging AI research areas, guiding future funding strategies.

Data Migration Consultant | GardaWorld (Montreal, Canada) *Sep. 2023 - Aug. 2024*
in collaboration with the Canadian Air Transport Security Authority

- Executed the migration of the legacy Airport Security Screening HR ecosystem to a new software suite across all 41 Canadian airports, achieving 100% data accuracy and zero downtime.
- Automated data migration scripts for employee records, schedules, and payroll information, reducing manual effort by 60% and ensuring seamless integration.
- Built linear regression models to predict airport traffic, improving scheduling efficiency by 25%.

AI Research Fellow | AI4Good Lab (Montreal, Canada) *May 2023 - Jul. 2023*

- Trained and tested a zero-shot model to classify AI projects into 6 categories targeted by Canadian AI regulations.
- Implemented **AITHICAL**, a prototype ML platform that aggregates AI regulations and guidelines, and suggests relevant sections of regulations to AI practitioners based on their projects descriptions.
- Developed a mentorship program to create a more inclusive and diverse Tech culture in Canada.

AI Analyst | Intact Insurance - Data Lab (Montreal, Canada) *May 2021 - Apr. 2023*

- Coordinated the development and deployment of XGBoost regression models to predict insurance premiums for 500K+ commercial and personal policies, achieving 98% model accuracy.
- Deployed NLP solutions to summarize insurance claims, enhancing efficiency for 10k+ employees.
- Conducted A/B testing, model validation, and user acceptance testing ensuring 95% user satisfaction rate.

AI Analyst | National Bank of Canada (Montreal, Canada) *Jun. 2018 - Apr. 2021*

- Implemented a Microsoft Azure chatbot for HR inquiries, serving 5K+ employees across Canada.
- Gathered and managed requirements for a smart search engine, enhancing data retrieval accuracy by 15%.
- Translated AI research findings into actionable insights aligned with organizational digitalization goals.

TEACHING AND TUTORING EXPERIENCE

University of Quebec in Montreal

- **Teaching Assistant** | INF 2050: Software Development Tools and Practices *Jan. 2019 - May 2019*
- **Teaching Assistant** | INF 1120: Introduction to Programming *Sep. 2018 - Dec. 2018*
- **Undergraduate students Tutor** | Software engineering courses *Jan. 2017 - Dec. 2017*

SELECTED AWARD & HONOR (BY YEAR)

- **Graduate Fellowship** | Penn State University *2024*
- **Highest Honors Master Thesis** | University of Quebec in Montreal *2020*
- **Faculty of Sciences Academic Excellence Scholarship** | University of Quebec in Montreal *2018*
- **Merit Scholarship for high-achieving exchange student** | University of Quebec in Montreal *2014*

PROFESSIONAL SERVICE

Student Affiliate at the Center for Socially Responsible Artificial Intelligence *Sep. 2024 - Now*
Engaging researchers from diverse disciplines in topics related to socially responsible AI.

Youth Mentor at Infinite Family, Not-for-profit organization *Jan. 2023 - Feb. 2024*
Mentoring high school students: tutoring on school subjects and advising on career paths.

Fundraising project manager at We the World, Not-for-profit organization *Jun. 2021 - Dec. 2022*
Coordinating online fundraising campaigns for mental health initiatives.